单液桥模型弯液面演化的数值模拟

Liskell Leo

liskello_o@outlook.com

❶ 研究背景

❷ 研究方法

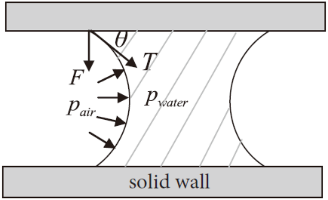
❸ 结果讨论

❹ 应用拓展

研究对象

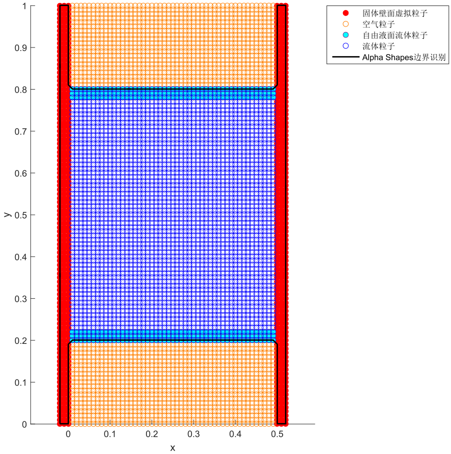
单液桥模型

两个平行固壁之间的弯液面



液桥力主要由表面张力与拉普拉斯压差引起!

θ：固-液静态接触角
F：附壁力
T：表面张力
p：压强



MATLAB 数值算例，初始的计算域被离散为2989个流体粒子、606个固体壁面虚构粒子和1960个空气粒子

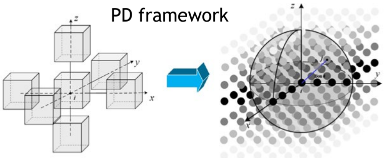
研究现状

有关弯液面的数值建模方法

研究方法	模型	特点	局限性
DDA	一般模型	固体被离散为不规则的多边形颗粒	弯液面的固液接触角和曲率半径恒定，不能获取动态演化过程
	二维扩展模型	根据固体单位体积的含水量确定弯液面形态	
DEM	一般模型	表面张力的作用由黏附力代替	缺乏对弯液面的描述
	孤立毛细管模型	适用于低饱和度的二维固体	引入了众多参数来描述弯液面的形态，可能带来较大的计算误差，且不能获取动态演化过程
	索道液桥模型	适用于高饱和度的二维固体	
	复杂流体团簇模型	适用于高饱和度的三维固体	
CFD-DEM	单向流固耦合模型	采用CFD方法模拟水压，采用DEM方法模拟固体颗粒运动，将基质吸力视为恒定的粘附力	流场不受固体颗粒运动的影响，不能完整地反映表面张力作用下的流固相互作用
MD	一般模型	纳米尺度的液桥模型	基于分子力场进行建模和计算，缺乏对表面张力的描述

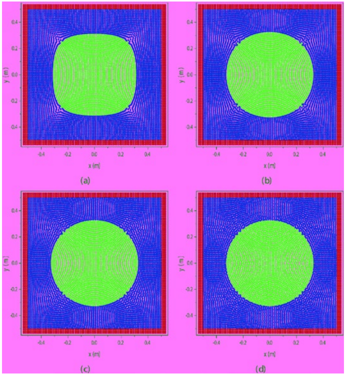
近场动力学在流固耦合研究中的应用

PD framework



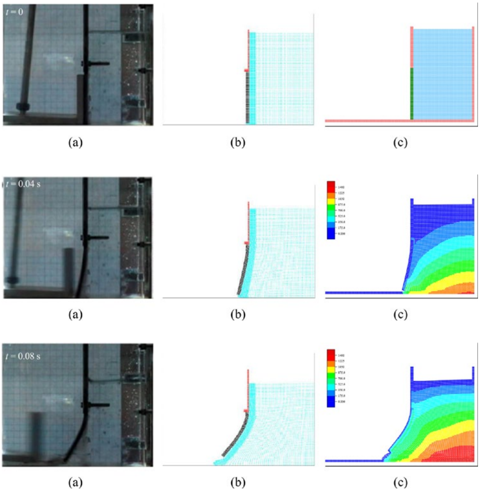
(Silling, 2000)

2D square droplet deformation simulation



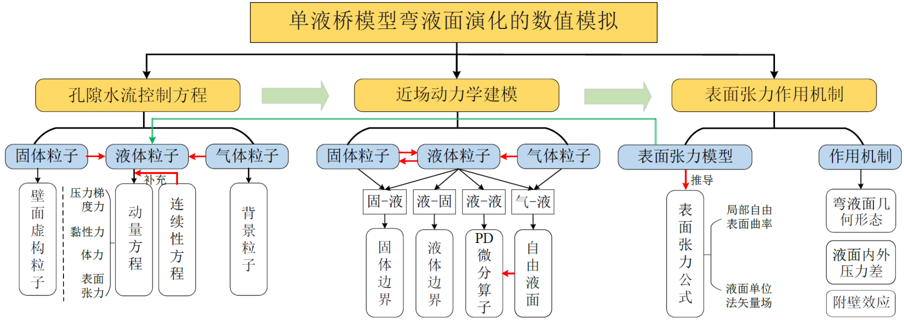
(Gao & Oterkus, 2020a)

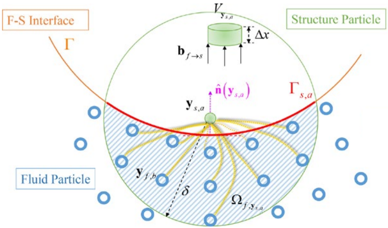
Dam collapse under an elastic rubber gate

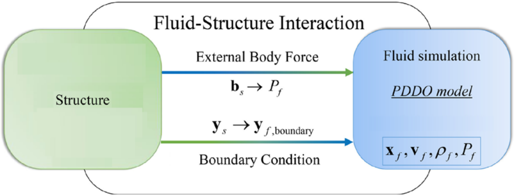


(Gao & Oterkus, 2020b)

技术路线



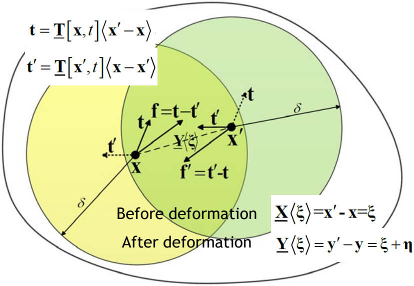




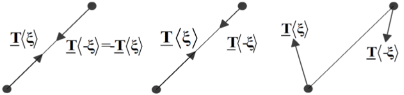
难点：计算复杂度高、规模大，边界条件施加困难！

近场动力学理论框架和微分算子

Interaction between material points in PD modeling framework



(Silling, 2008)



Bond-based　　Ordinary state-based　　Unordinary state-based

Peridynamic differential operators

$$\frac{\partial^{n_1+n_2+\cdots n_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \cdots \partial x_M^{p_M}} = \int_{H_\mathbf{x}} f(\mathbf{x}+\boldsymbol{\xi}) g_N^{p_1 p_2 \cdots p_N}(\boldsymbol{\xi}) dV_{\mathbf{x}'}$$

(Madenci et al., 2016)

Governing equations of free surface flow $\xrightarrow{\text{PDDO}}$ Non-local form expression

Lagrangian description

$$\frac{\partial \rho}{\partial t} = -\rho \nabla \cdot \mathbf{v}$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} = -\nabla P + \mathbf{F_V} + \mathbf{F_B} + \mathbf{F_S}$$

where

$$P = \frac{\rho_0 c_f^2}{\gamma}\left(\left(\frac{\rho}{\rho_0}\right)^{\gamma} - 1\right) + P_0$$

$$\mathbf{F_V} = \mu(\Delta \mathbf{v})$$

$$\mathbf{F_B} = 0$$

$$\mathbf{F_S} = \beta \kappa \mathbf{n} \nabla s$$

PD integral equations

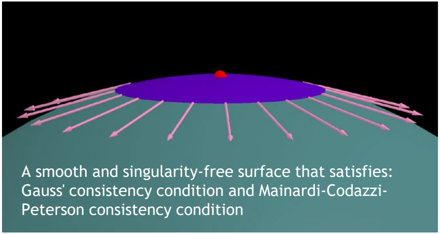
$$\frac{\partial \rho(\mathbf{x})}{\partial t} = -\rho(\mathbf{x}) \int_{H_\mathbf{x}} \mathbf{g}_1(\boldsymbol{\xi}) \cdot (\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) dV_{\mathbf{x}'}$$

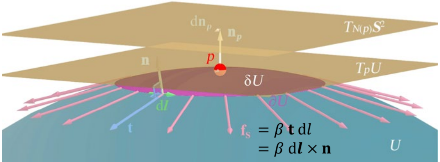
$$\rho \frac{\partial \mathbf{v}}{\partial t} = -\int_{H_\mathbf{x}} (P(\mathbf{x}') - P(\mathbf{x})) \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'} + \int_{H_\mathbf{x}} \mu(\mathbf{x})(\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) \mathrm{tr}(\mathbf{g}_2(\boldsymbol{\xi})) dV_{\mathbf{x}'} - \beta \int_{H_\mathbf{x}} \kappa (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'}$$

where

$$\kappa = \frac{\int_{H_\mathbf{x}} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'}}{\int_{H_\mathbf{x}} w_0(\|\boldsymbol{\xi}\|) dV_{\mathbf{x}'} / \int_{H_\mathbf{x}} w_0(\|\boldsymbol{\xi}\|) dV_{\mathbf{x}'}}$$

界面流体元的表面张力模型





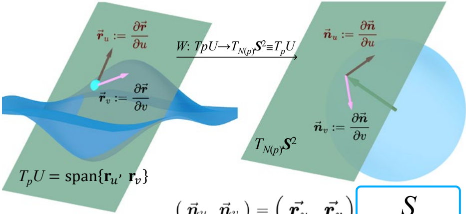
$$\mathbf{f}_s = \beta\,\mathbf{t}\,\mathrm{d}l = \beta\,\mathrm{d}\boldsymbol{l}\times\mathbf{n}$$

$$\delta\mathbf{F_S} = \oint_{\partial U}\beta \mathrm{d}\boldsymbol{l}\times\mathbf{n} = \oint_{\partial U}\beta \mathrm{d}\boldsymbol{l}\times\mathbf{n}_p + \boxed{\oint_{\partial U}\beta \mathrm{d}\boldsymbol{l}\times\mathrm{d}\mathbf{n}_p}$$
$$= -\mathbf{n}_p\oint_{\partial U}(\beta\mathrm{d}\mathbf{n}_p)\cdot(\mathbf{t}\,\mathrm{d}l)$$

Gauss theorem $\Rightarrow \mathbf{n}_p\oint_{\partial U}(\beta\mathrm{d}\mathbf{n}_p)\cdot(\mathbf{t}\,\mathrm{d}l) = \beta(\nabla\cdot\mathrm{d}\mathbf{n}_p)\delta U\,\mathbf{n}_p$

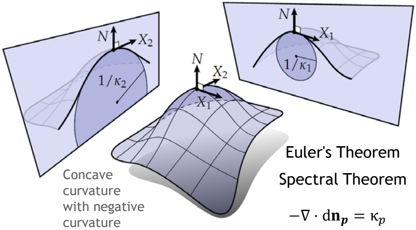
$$\mathbf{F_S} = \lim_{\delta U\to 0}\frac{\delta\mathbf{F_S}}{\delta U} = \beta\left(-\nabla\cdot\mathrm{d}\mathbf{n}_p\right)\mathbf{n}_p$$

$$\vec{r}(u,v) = \begin{pmatrix} x(u,v)\\ y(u,v)\\ z(u,v)\end{pmatrix} \xrightarrow{N:\,U\to S^2\subset\mathbf{R}^3} \frac{\vec{r}_u\times\vec{r}_v}{\|\vec{r}_u\times\vec{r}_v\|} = \boxed{\vec{n}(u,v)} = \begin{pmatrix} n_1(u,v)\\ n_2(u,v)\\ n_3(u,v)\end{pmatrix}$$



$$\begin{pmatrix}\vec{n}_u & \vec{n}_v\end{pmatrix} = \begin{pmatrix}\vec{r}_u & \vec{r}_v\end{pmatrix}\boxed{S}$$

Shape Matrix



Euler's Theorem

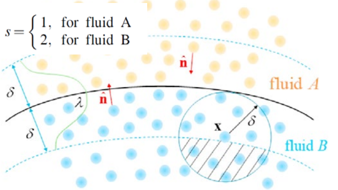
Spectral Theorem

$$-\nabla\cdot\mathrm{d}\mathbf{n}_p = \kappa_p$$

表面张力计算、控制方程、时间差分格式

The non-local form of the surface tension formula

$$s=\begin{cases}1, & \text{for fluid A}\\ 2, & \text{for fluid B}\end{cases}$$



$$(s(x')-s(\mathbf{x}))=\mathbb{C}(x',\mathbf{x})=\begin{cases}\dfrac{2\rho_{x,0}}{\rho_{x,0}+\rho_{x',0}}, & \text{if } \mathbf{x} \text{ and } x' \text{ belong to different phases}\\ 0, & \text{if } \mathbf{x} \text{ and } x' \text{ belong to the same phase}\end{cases}$$

Introduce the truncation function to avoid the unit normal vector field having incorrect directions

$$\mathbf{n}(\mathbf{x})=\begin{cases}\dfrac{\int_{H_\mathbf{x}}(s(\mathbf{x}')-s(\mathbf{x}))\mathbf{g}_1(\xi)\mathrm{d}V_{\mathbf{x}'}}{\left\|\int_{H_\mathbf{x}}(s(\mathbf{x}')-s(\mathbf{x}))\mathbf{g}_1(\xi)\mathrm{d}V_{\mathbf{x}'}\right\|}, & \text{if } \|\nabla s\|>1.0\times10^{-2}\Delta x\\ 0, & \text{else}\end{cases}$$

$$\kappa^*(\mathbf{x})=-\nabla\cdot\mathbf{n}(\mathbf{x})=-\int_{H_\mathbf{x}}(\mathbf{n}(\mathbf{x}')-\mathbf{n}(\mathbf{x}))\cdot\mathbf{g}_1(\xi)\mathrm{d}V_{\mathbf{x}'}$$

Introduce correction factors to correct the curvature field of boundary particles

$$\text{let } \mathbb{N}(\mathbf{x})=\begin{cases}1, & \text{if } \|\nabla s\|>1.0\times10^{-2}\Delta x\\ 0, & \text{else}\end{cases}$$

$$\varsigma^*(\mathbf{x})=\frac{\int_{H_\mathbf{x}}\min(\mathbb{N}(\mathbf{x}),\mathbb{N}(\mathbf{x}'))w_0(\|\boldsymbol{\xi}\|)\mathrm{d}V_{\mathbf{x}'}}{\int_{H_\mathbf{x}}w_0(\|\boldsymbol{\xi}\|)\mathrm{d}V_{\mathbf{x}'}},\quad \kappa^{**}(\mathbf{x})=\frac{\kappa^*(\mathbf{x})}{\varsigma^*(\mathbf{x})}$$

(Gao & Oterkus, 2020a)

- **Continuity Equation**

$$\rho_i^{n+1}=\rho_i^n-\rho_i^n\Delta t\sum_{j=1}^{N_i}\left[(\mathbf{v}_j^n-\mathbf{v}_i^n)\cdot\mathbf{g}_1(\boldsymbol{\xi}_{ij}^n)\right]V_j$$

$$\rho_i^{n+1}=\rho_i^n-\frac{2\Delta x^2}{\pi\delta^2}\rho_i^n\Delta t\sum_{j=1}^{N_i}(\mathbf{v}_j^n-\mathbf{v}_i^n)\frac{[\xi_{ij1}\quad \xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}$$

For two-dimensional problems, the relative value form of the near-field dynamic differential operator is adopted.

- **Momentum equation**

$$\mathbf{a}_i^{n+1}=\frac{1}{\rho_i^{n+1}}\left(\begin{array}{c}-\sum_{j=1}^{N_i}(P_j^n-P_i^n)\mathbf{g}_1(\boldsymbol{\xi}_{ij}^n)V_j+\sum_{j=1}^{N_i}\mu_i(\mathbf{v}_j^n-\mathbf{v}_i^n)\mathrm{tr}\left(\mathbf{g}_2(\boldsymbol{\xi}_{ij}^n)\right)V_j\\ -\beta\dfrac{\left(\sum_{j=1}^{N_i}\min(\mathbb{N}_i,\mathbb{N}_j)(\mathbf{n}_j^n-\mathbf{n}_i^n)\cdot\mathbf{g}_1(\boldsymbol{\xi}_{ij}^n)V_j\right)}{\sum_{j=1}^{N_i}\min(\mathbb{N}_i,\mathbb{N}_j)w_0(\|\boldsymbol{\xi}_{ij}^n\|)V_j/\sum_{j=1}^{N_i}w_0(\|\boldsymbol{\xi}_{ij}^n\|)V_j}\left(\sum_{j=1}^{N_i}(s(\mathbf{x}')-s(\mathbf{x}))\mathbf{g}_1(\boldsymbol{\xi}_{ij}^n)V_j\right)\end{array}\right)+\frac{1}{\rho_i^{n+1}\mathbf{F}^B}$$

$$\mathbf{a}_i^{n+1}=\frac{2\Delta x^2}{\pi\delta^2\rho_i^{n+1}}\left(\begin{array}{c}-\sum_{j=1}^{N_i}(P_j^n-P_i^n)\dfrac{[\xi_{ij1}\quad\xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}+\dfrac{3\mu_i}{\delta}\sum_{j=1}^{N_i}\dfrac{(\mathbf{v}_j^n-\mathbf{v}_i^n)}{\sqrt{\|\xi_{ij1}^2+\xi_{ij2}^2\|}}\\ -\dfrac{\left(\sum_{j=1}^{N_i}\min(\mathbb{N}_i,\mathbb{N}_j)(\mathbf{n}_j^n-\mathbf{n}_i^n)\dfrac{[\xi_{ij1}\quad\xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}\right)\left(\sum_{j=1}^{N_i}(s(\mathbf{x}')-s(\mathbf{x}))\dfrac{[\xi_{ij1}\quad\xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}\right)}{\dfrac{\pi\delta^2}{2\beta\Delta x^2}\left(\sum_{j=1}^{N_i}\min(\mathbb{N}_i,\mathbb{N}_j)w_0(\|\boldsymbol{\xi}_{ij}^n\|)V_j/\sum_{j=1}^{N_i}w_0(\|\boldsymbol{\xi}_{ij}^n\|)V_j\right)}\end{array}\right)+\frac{1}{\rho_i^{n+1}\mathbf{F}^B}$$

- **Time difference scheme**

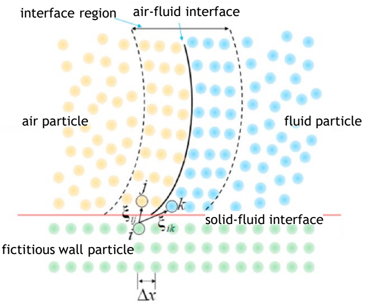
Second-order Verlet-Velocity algorithm

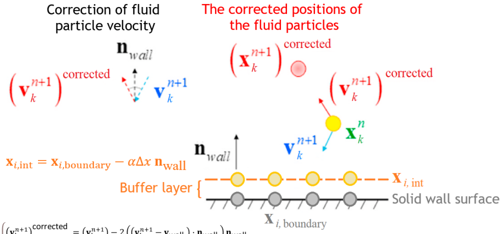
$$\begin{cases}\mathbf{v}_i^{n+1}=\mathbf{v}_i^n+\dfrac{1}{2}(\mathbf{a}_i^{n+1}+\mathbf{a}_i^n)\Delta t\\ \mathbf{u}_i^{n+1}=\mathbf{u}_i^n+\mathbf{v}_i^n\Delta t+\dfrac{1}{2}\mathbf{a}_i^n\Delta t^2\\ \mathbf{x}_i^{n+1}=\mathbf{x}_i^0+\mathbf{u}_i^{n+1}\end{cases}\qquad \mathbf{u}_i^0$$

Courant-Friedrichs-Levy stability condition

$$\Delta t<\min\left(\frac{h_{\mathrm{smooth}}}{|\mathbf{v}_{\max}|+c_B},\frac{1}{8}\frac{h_{\mathrm{smooth}}{}^2}{\max(\mu/\rho_0)},\frac{1}{4}\sqrt{\frac{h_{\mathrm{smooth}}}{g}},\frac{1}{4}\sqrt{\frac{\min(\rho_0)h^3}{2\pi\beta}}\right)$$

边界的处理





$\mathbf{x}_{i,\text{int}} = \mathbf{x}_{i,\text{boundary}} - \alpha\Delta x\, \mathbf{n}_{\text{wall}}$

$$\begin{cases} \left(\mathbf{v}_k^{n+1}\right)^{\text{corrected}} = \left(\mathbf{v}_k^{n+1}\right) - 2\left(\left(\mathbf{v}_k^{n+1} - \mathbf{v}_{\text{wall}}\right)\cdot \mathbf{n}_{\text{wall}}\right)\mathbf{n}_{\text{wall}} \\ \left(\mathbf{u}_k^{n+1}\right)^{\text{corrected}} = \mathbf{u}_k^{n} + \left(\mathbf{v}_k^{n+1}\right)^{\text{corrected}}\Delta t + \frac{1}{2}\mathbf{a}_k^n \Delta t^2 \\ \left(\mathbf{x}_k^{n+1}\right)^{\text{corrected}} = \mathbf{x}_k^{n} + \left(\mathbf{u}_k^{n+1}\right)^{\text{corrected}} \end{cases}$$

Fluid particles that penetrate and enter the interior of the solid wall surface

Criterion: $\left(\mathbf{x}_k^{n+1} - \mathbf{x}_{i,\text{int}}\right)\cdot \mathbf{n}_{\text{wall}} < 0$

Considering the no-slip boundary, the fluid velocity is zero at the wall surface.

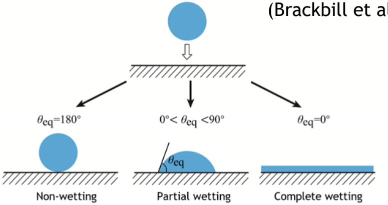
$$\mathbf{v}_i^n = 2\mathbf{v}_{\text{wall}} - \sum_{k=1}^{N_f} w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)\mathbf{v}_k^n$$

$$P_i = \frac{\sum_{k=1}^{N_f}\left(P_k + \left(\mathbf{F}_k^B - \rho_k \mathbf{a}_{\text{wall}}\right)\cdot \boldsymbol{\xi}_{ik}^n\right) w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)}{\sum_{k=1}^{N_f} w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)}$$

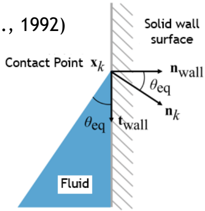
$$\rho_i = \rho_{0,i}\left(\frac{P_i - P_{0,i}}{P_{ref,i}} + 1\right)^{1/\gamma_i}$$

(Gao & Oterkus, 2020a)

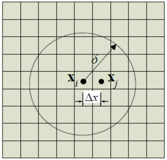
Consider the surface properties of the solid



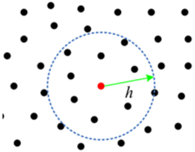
(Brackbill et al., 1992)



空间离散（近场域搜索）和自由液面识别

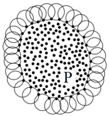


Two-dimensional structure with central orthogonal uniform discretization.

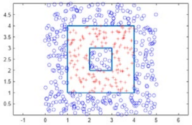


By calling the library function rangesearch to conduct a KD-tree neighborhood search for the specified particle.

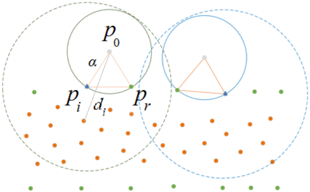
The free surface is identified by calling library functions such as:



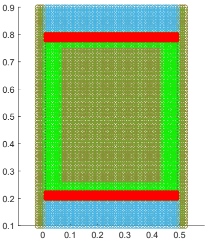
alphaShape

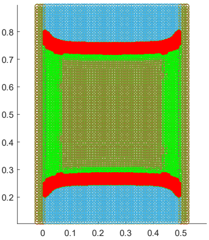


inpolygon



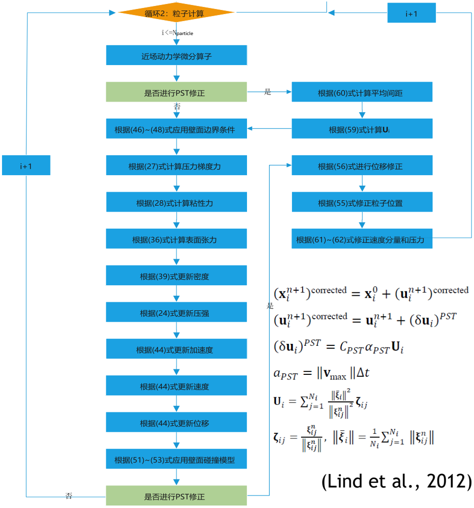
对所有流体粒子和固体粒子构成的结构使用alpha shapes算法，可以找到该结构的边界点。再使用inpolygon函数剔除背景中多余空气粒子！





防聚集的粒子迁移技术

- PST（Particle Shifting Technology）技术通过引入位移修正来调整粒子位置，使粒子在移动后保持近似均匀分布



$$(\mathbf{x}_i^{n+1})^{\text{corrected}} = \mathbf{x}_i^0 + (\mathbf{u}_i^{n+1})^{\text{corrected}}$$

$$(\mathbf{u}_i^{n+1})^{\text{corrected}} = \mathbf{u}_i^{n+1} + (\delta\mathbf{u}_i)^{PST}$$

$$(\delta\mathbf{u}_i)^{PST} = C_{PST}\alpha_{PST}\mathbf{U}_i$$

$$\alpha_{PST} = \|\mathbf{v}_{\max}\|\Delta t$$

$$\mathbf{U}_i = \sum_{j=1}^{N_i}\frac{\|\bar{\xi}_i\|^2}{\|\xi_{ij}^n\|^2}\zeta_{ij}$$

$$\zeta_{ij} = \frac{\xi_{ij}^n}{\|\xi_{ij}^n\|},\quad \|\bar{\xi}_i\| = \frac{1}{N_i}\sum_{j=1}^{N_i}\|\xi_{ij}^n\|$$

(Lind et al., 2012)

$$\begin{aligned}(\mathrm{v}_i^{n+1})^{\text{corrected}} = \Bigg[\; & \mathrm{v}_{i1}^{n+1} + \frac{2\Delta x^2}{\pi\delta^2}\sum_{j=1}^{N_i}(\mathrm{v}_{j1}^n-\mathrm{v}_{i1}^n)\frac{[\xi_{ij1}\ \ \xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}\cdot(\delta\mathbf{u}_i)^{PST} \\ & + \frac{3\Delta x^2}{2\pi\delta^3}\sum_{j=1}^{N_i}((\delta\mathbf{u}_i)^{PST})^{\mathrm{T}}\cdot(\mathrm{v}_{j1}^n-\mathrm{v}_{i1}^n)\frac{\begin{pmatrix}3\xi_1^2-\xi_2^2 & 4\xi_1\xi_2\\ 4\xi_1\xi_2 & -\xi_1^2+3\xi_2^2\end{pmatrix}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|^{3/2}}\cdot(\delta\mathbf{u}_i)^{PST}, \\ & \mathrm{v}_{i2}^{n+1} + \frac{2\Delta x^2}{\pi\delta^2}\sum_{j=1}^{N_i}(\mathrm{v}_{j2}^n-\mathrm{v}_{i2}^n)\frac{[\xi_{ij1}\ \ \xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}\cdot(\delta\mathbf{u}_i)^{PST} \\ & + \frac{3\Delta x^2}{2\pi\delta^3}\sum_{j=1}^{N_i}((\delta\mathbf{u}_i)^{PST})^{\mathrm{T}}\cdot(\mathrm{v}_{j2}^n-\mathrm{v}_{i2}^n)\frac{\begin{pmatrix}3\xi_1^2-\xi_2^2 & 4\xi_1\xi_2\\ 4\xi_1\xi_2 & -\xi_1^2+3\xi_2^2\end{pmatrix}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|^{3/2}}\cdot(\delta\mathbf{u}_i)^{PST}\;\Bigg]^{\mathrm{T}}\end{aligned}$$

$$\begin{aligned}(P_i^{n+1})^{\text{corrected}} = & P_i^{n+1} + \frac{2\Delta x^2}{\pi\delta^2}\sum_{j=1}^{N_i}(P_j^n-P_i^n)\frac{[\xi_{ij1}\ \ \xi_{ij2}]^{\mathrm{T}}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|}\cdot(\delta\mathbf{u}_i)^{PST} \\ & + \frac{3\Delta x^2}{2\pi\delta^3}\sum_{j=1}^{N_i}((\delta\mathbf{u}_i)^{PST})^{\mathrm{T}}\cdot(P_j^n-P_i^n)\frac{\begin{pmatrix}3\xi_1^2-\xi_2^2 & 4\xi_1\xi_2\\ 4\xi_1\xi_2 & -\xi_1^2+3\xi_2^2\end{pmatrix}}{\|\xi_{ij1}^2+\xi_{ij2}^2\|^{3/2}}\cdot(\delta\mathbf{u}_i)^{PST}\end{aligned}$$

速度和压力也可以根据位移的修正量进行修正
(二维完整对称近场域下相对值形式)

(Gao & Oterkus, 2020a)

❶ 研究背景

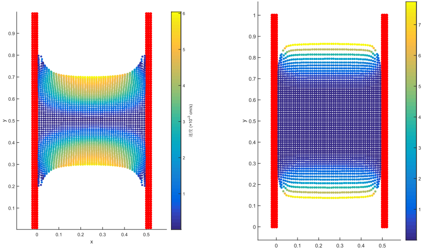
❷ 研究方法

❸ 结果讨论

❹ 应用拓展

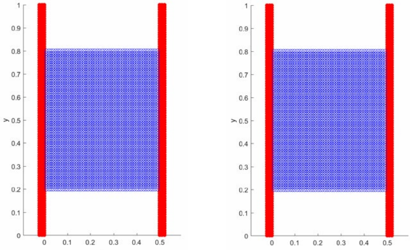
计算结果

Final configuration



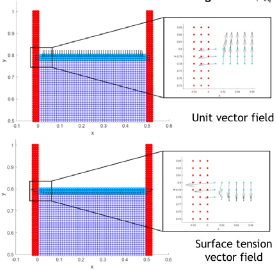
(a) $\theta_{eq} = 5°$ (b) $\theta_{eq} = 175°$

Liquid level evolution process



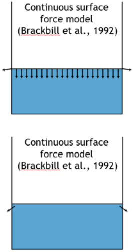
(a) $\theta_{eq} = 5°$ (b) $\theta_{eq} = 175°$

Initial configuration($\theta_{eq} = 5°$)

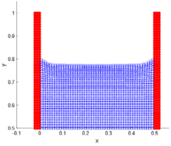


Unit vector field

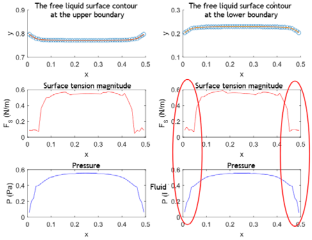
Surface tension vector field



Particle distribution at 7500 time steps for the concave liquid surface example ($\theta_{eq} = 5°$)

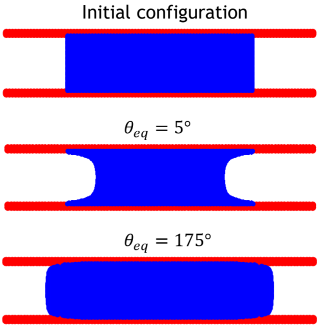


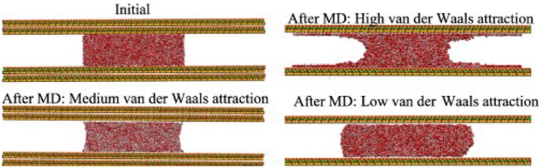
Note that the absolute value of the component of surface tension along the y-axis is selected to replace the surface tension for analysis.



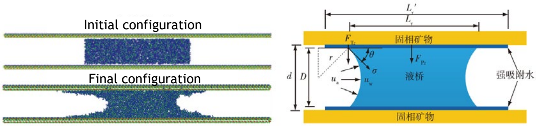
算例比较（PD vs. MD）

The stronger the adsorption of the solid boundary on the pore water, the more "concave" the curved liquid surface will be; conversely, it will be more "convex".





(Amarasinghe et al., 2014)



(Luo et al., 2022)

(a) PD results

(b) MD results

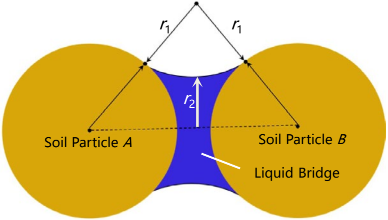
❶ 研究背景

❷ 研究方法

❸ 结果讨论

❹ 应用拓展

非饱和黏土弯液面演化的数值模拟

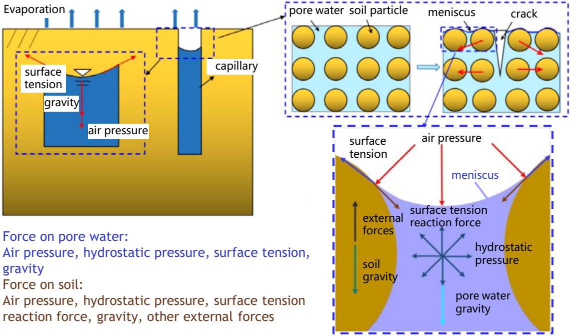


The matrix suction of unsaturated soil is defined by Young-Laplace Equation, related to the morphology curved liquid surface

$$s = \gamma_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

找到黏土干缩开裂与微观液桥结构的联系是关键！

Micro-scale meniscus evolution controls capillary suction → governs macro-scale shrinkage and cracking!



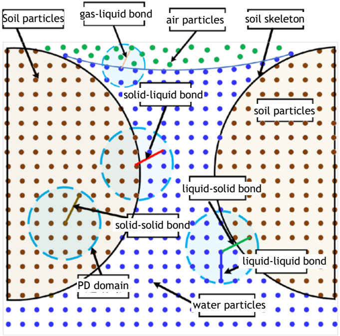
Force on pore water:
Air pressure, hydrostatic pressure, surface tension, gravity
Force on soil:
Air pressure, hydrostatic pressure, surface tension reaction force, gravity, other external forces

“热-水-力”耦合作用下弯液面的演化探究

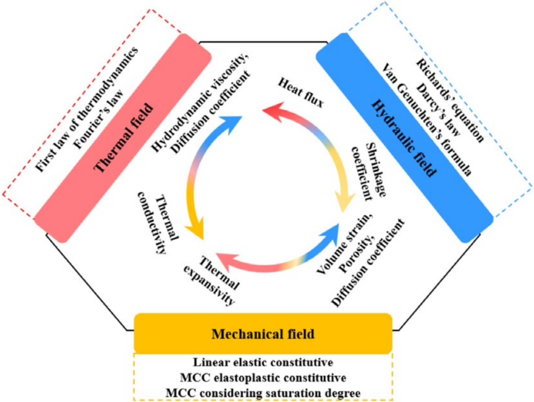
Simplified single liquid bridge model

Capillary action ✓
Evaporation effect ✗

Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!



The microscopic structure of clay

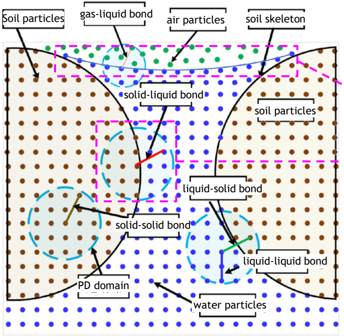


“热-水-力”耦合作用下弯液面的演化探究

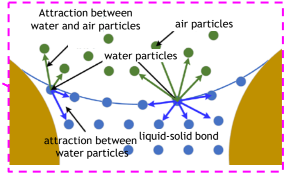
Simplified single liquid bridge model

Capillary action ✓
Evaporation effect ✗

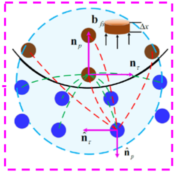
Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!



The microscopic structure of clay



One-way gas-liquid interaction



Fluid-solid interaction

Boundary conditions:

Gas → Liquid: Surface tension
Solid ⇄ Liquid: Decomposes into two forces acting in the pressure direction and the shear direction

“热-水-力”耦合作用下的数学物理建模

The basic equations of the micro-scale "solid-liquid-gas" three-phase medium of clay under the "thermal-hydrodynamic" coupling effect:

mass conservation (continuity) equation

$$\frac{D\rho_l(T)}{Dt} + \rho_l(T)\nabla\cdot\mathbf{U}_l = \rho_l(T)\dot{m}\left(\frac{1}{\rho_g(T)} - \frac{1}{\rho_l(T)}\right) \qquad \frac{D\rho_s}{Dt} + \rho_s\nabla\cdot\mathbf{U}_l = 0$$

momentum conservation equation

$$\frac{D\left(\rho_l(T)\mathbf{U}_l\right)}{Dt} = -\nabla p(T) + \mathbf{F}_l^v + \mathbf{F}_l^B + \mathbf{F}_l^{sl} \qquad m_s\frac{\partial\mathbf{U}_s}{\partial t} = \mathbf{F}_s^{pv} + \mathbf{F}_s^B + \mathbf{F}_s^{ls} + \mathbf{F}_s^S$$

water vapor diffusion equation

$$\frac{\partial Y}{\partial t} + \nabla\cdot\left(Y\mathbf{U}_l\right) = \nabla\cdot\left(D_v(T)\nabla Y\right)$$

heat energy conservation equation

$$\frac{\partial\left(\rho_l c_{lp} T_l\right)}{\partial t} + \nabla\cdot\left(\rho_l c_{lp}\mathbf{U}_l T_l\right) = \nabla\cdot\left(k_{lT}\nabla T_l\right) - \dot{m}h_{ev} + \frac{1}{2}\mu(T_l)\mathbf{S}:\mathbf{S}$$

heat transfer equation

conduction

$$\frac{\partial\left(\rho_s c_{sp} T_s\right)}{\partial t} = \nabla\cdot\left(k_{sT}\nabla T_s\right)$$

convection

$$k_{sT}\frac{\partial T_s}{\partial n_{sl}} = \alpha_{sl}\left(T_s - T_l\right)$$

谢谢！

Thanks for your time and attention!